

A fractional Trudinger upgrade for Orlicz–Besov extension domains

Candidate partial-result packet for arXiv:1901.06186

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Status. Candidate substantial partial result, likely valid, pending human review.

Source problem. Liang and Zhou [1] prove that a homogeneous Orlicz–Besov extension domain is Ahlfors n -regular when its Young function is subexponential at infinity. Remark 1.3 on source PDF page 3 conjectures that the subexponential hypothesis is unnecessary.

ORLICZ-BESOV EXTENSION AND AHLFORS n -REGULAR DOMAINS

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As a byproduct, we have the following result.

Corollary 1.2. *Suppose that ϕ is a Young function satisfying (1.2) and (1.3). Let $\Omega \subset \mathbb{R}^n$ be any domain. The following are equivalent:*

- (i) Ω is Ahlfors n -regular;
- (ii) Ω is a $\dot{\mathbf{B}}^\phi$ -extension domain;
- (iii) Ω supports the imbedding (1.5).

Remark 1.3. We conjecture that Theorem 1.1 (ii) holds without the additional assumption (1.3). The difficulty to remove (1.3) is to find a suitable imbedding properties of $\dot{\mathbf{B}}^\phi(\mathbb{R}^n)$ better than (1.5) when ϕ does not satisfy (1.3).

Note that (1.5) is always true when Ω is a $\dot{\mathbf{B}}^\phi$ -extension domain, but it is not enough to prove that Ω is Ahlfors n -regular in general. If $\phi(t) = e^{t^\alpha} - \sum_{j=0}^{\lfloor n/\alpha \rfloor} t^{\gamma j} / j!$ for $t \geq 0$, by Lemma 2.5, Ω supports the imbedding

$$\int_B \exp\left(\frac{|u(x) - u_B|}{\alpha}\right) dx \leq C(n)$$

whenever $\alpha > C(\gamma, n)\|u\|_{\dot{\mathbf{B}}^{\phi_\gamma}(\mathbb{R}^n)}$. However, when $\dot{\mathbf{B}}^\phi$ -extension domain, such an imbedding is also not enough to prove Ω is Ahlfors n -regular.

Notation used in the following is standard. The constant $C(n, \alpha, \phi)$ would vary from line to line and is

Result and proof intuition

The source uses the embedding $\dot{\mathbf{B}}^\phi \hookrightarrow \text{BMO}$ and John–Nirenberg, which gives only exponential-linear integrability. If ϕ dominates a power t^p with $p > n$, then $\dot{\mathbf{B}}^\phi$ instead controls the critical fractional space $W^{n/p, p}$. Fractional Moser–Trudinger raises the exponential power from 1 to $p' = p/(p-1)$. In the source’s density iteration, this replaces $\phi(C \log q)$ by $\phi(C(\log q)^{1/p'})$, allowing growth almost as fast as $\exp(t^{p'})$.

For a Young function ϕ , write

$$\|u\|_{\dot{\mathbf{B}}^\phi(U)} = \inf \left\{ \lambda > 0 : \int_U \int_U \phi\left(\frac{|u(x) - u(y)|}{\lambda}\right) \frac{dx dy}{|x - y|^{2n}} \leq 1 \right\}.$$

Theorem 1. *Let $n \geq 2$, and let ϕ satisfy the source’s nontriviality condition*

$$C_\phi = \sup_{t>0} \frac{t^n}{\phi(t)} \int_0^t \frac{\phi(s)}{s^{n+1}} ds < \infty.$$

Suppose that for some $p > n$ and $a > 0$,

$$\phi(t) \geq at^p \quad (t \geq 0), \quad (1)$$

and, with $p' = p/(p-1)$,

$$\lim_{t \rightarrow \infty} \frac{\log(1 + \phi(t))}{t^{p'}} = 0. \quad (2)$$

If a domain $\Omega \subset \mathbb{R}^n$ is a $\dot{\mathbf{B}}^\phi$ -extension domain, then it is Ahlfors n -regular.

Condition (2) is strictly weaker at infinity than the source's $\log(1 + \phi(t)) = o(t)$, because $p' > 1$.

The stronger local embedding

For $s = n/p \in (0, 1)$, use

$$[v]_{s,p}^p = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \frac{|v(x) - v(y)|^p}{|x - y|^{2n}} dx dy.$$

Lemma 1 (Localized fractional Trudinger). *There are $c, C > 0$, depending only on n, p , such that for every ball $B \subset \mathbb{R}^n$ and every $v \in L_{\text{loc}}^p(\mathbb{R}^n)$ with $[v]_{s,p} < \infty$, one can choose a constant $c_B(v)$ for which*

$$\frac{1}{|B|} \int_B \exp \left[c \left(\frac{|v - c_B(v)|}{[v]_{s,p}} \right)^{p'} \right] dx \leq C. \quad (3)$$

The assertion is interpreted trivially when $[v]_{s,p} = 0$.

Proof. Let $2B$ be the concentric doubled ball, put $c_B(v) = v_{2B}$, and choose a Lipschitz cutoff η equal to 1 on B and supported in $2B$, with $|\nabla \eta| \lesssim r(B)^{-1}$. For $w = \eta(v - v_{2B})$, split the seminorm integral into pairs in $2B \times 2B$ and pairs with exactly one point in $2B$. The product decomposition on the first part, the tail-kernel bound on the second, and fractional Poincaré give

$$[w]_{s,p}^p \leq C[n, p] ([v]_{s,p}^p + r^{-n} \|v - v_{2B}\|_{L^p(2B)}^p) \leq C[n, p] [v]_{s,p}^p.$$

Here the cutoff and tail terms are bounded using

$$\int_{\mathbb{R}^n} \frac{\min\{1, (|x - y|/r)^p\}}{|x - y|^{2n}} dx \lesssim r^{-n}$$

and $\|v - v_{2B}\|_{L^p(2B)}^p \lesssim r^{sp} [v]_{s,p}^p = r^n [v]_{s,p}^p$. Now $w \in \widetilde{W}_0^{s,p}(2B)$. Proposition 3.1 of Parini–Ruf [2], followed by scaling, gives

$$\frac{1}{|2B|} \int_{2B} \exp \left[c_0 \left(\frac{|w|}{[w]_{s,p}} \right)^{p'} \right] dx \leq C_0.$$

Since $w = v - v_{2B}$ on B , reducing c_0 proves (3). □

3. VALIDITY OF MOSER-TRUDINGER INEQUALITY

In this section we give a short proof of the validity of the Moser-Trudinger inequality for some value of $\alpha > 0$, which is essentially contained in [19], following the approach by Trudinger.

Proposition 3.1. *Let Ω be a bounded, open domain of $\mathbb{R}^N$, and let $sp = N$. There exists $\alpha_* = \alpha_*(s, \Omega) > 0$ such that*

$$\sup \left\{ \int_{\Omega} \exp \left(\alpha |u|^{\frac{N}{N-s}} \right) \middle| u \in \widetilde{W}_0^{s,p}(\Omega), [u]_{W^{s,p}(\mathbb{R}^N)} \leq 1 \right\} < +\infty \quad \text{for } \alpha \in [0, \alpha_*).$$

Let $E : \dot{\mathbf{B}}^\phi(\Omega) \rightarrow \dot{\mathbf{B}}^\phi(\mathbb{R}^n)$ be an extension operator of norm K . By (1) and the Luxemburg definition,

$$[Eu]_{s,p} \leq a^{-1/p} \|Eu\|_{\dot{\mathbf{B}}^\phi(\mathbb{R}^n)} \leq a^{-1/p} K \|u\|_{\dot{\mathbf{B}}^\phi(\Omega)}. \quad (4)$$

Applying Lemma 1 to Eu and restricting to $B \cap \Omega$ therefore yields constants A, C such that

$$\inf_{c \in \mathbb{R}} \int_{B \cap \Omega} \exp \left[\left(\frac{|u - c|}{A \|u\|_{\dot{\mathbf{B}}^\phi(\Omega)}} \right)^{p'} \right] dx \leq C |B| \quad (5)$$

for every ball B and nonconstant $u \in \dot{\mathbf{B}}^\phi(\Omega)$.

Density iteration

Fix $x \in \Omega$ and a radius $r > 0$, and set

$$v = |B(x, r) \cap \Omega|, \quad \rho = \frac{v}{r^n}.$$

First suppose $r < \text{diam}(\Omega)/4$. Choose decreasing $b_0 = 1 > b_1 > b_2 > \dots \rightarrow 0$ so that

$$|B(x, b_j r) \cap \Omega| = 2^{-j} v. \quad (6)$$

For $j \geq 1$, let u_j be 1 on $B(x, b_{j+1} r) \cap \Omega$, linear in the radial variable on the next annulus, and 0 outside $B(x, b_j r) \cap \Omega$. Source Lemma 5.1 gives

$$\|u_j\|_{\dot{\mathbf{B}}^\phi(\Omega)} \leq \frac{C_0}{\phi^{-1}(A_j)}, \quad A_j = \frac{2^j (b_j - b_{j+1})^n}{\rho}. \quad (7)$$

Inside $B(x, b_{j-1} r) \cap \Omega$, the function is 1 on a set of measure $2^{-j-1} v$ and 0 on a disjoint set of at least that measure. For every $c \in \mathbb{R}$, one of these two sets satisfies $|u_j - c| \geq 1/2$. Hence (5) and (7) imply

$$2^{-j-1} v \exp \left[c_1 \phi^{-1}(A_j)^{p'} \right] \leq C r^n.$$

After taking logarithms and applying ϕ ,

$$A_j \leq \phi \left(C_1 [\log(C_2 2^j / \rho)]^{1/p'} \right). \quad (8)$$

By (2), choose the little- o constant so that the right side of (8) is at most $C(2^j / \rho)^{1/2}$ (enlarging C handles bounded arguments). Consequently

$$(b_j - b_{j+1})^n = 2^{-j} \rho A_j \leq C \rho^{1/2} 2^{-j/2}.$$

Summing the n th roots gives

$$b_1 \leq C \rho^{1/(2n)}. \quad (9)$$

If $b_1 \geq 1/10$, formula (9) immediately gives $\rho \geq c > 0$. If $b_1 < 1/10$, connectedness and $r < \text{diam}(\Omega)/4$ provide $x' \in \Omega$ with $|x - x'| = b_1 r + r/5$. Put $R = 2r/5$. Then

$$B(x, b_1 r) \subset B(x', R) \subset B(x, r), \quad B(x, b_1 r) \cap B(x', R/2) = \emptyset.$$

Since the first halving in (6) splits the volume of $B(x, r) \cap \Omega$ equally, one obtains

$$|B(x', R/2) \cap \Omega| \leq \frac{1}{2} |B(x', R) \cap \Omega|.$$

Thus the first halving radius for (x', R) is at least $1/2$, so the previous case applied at (x', R) gives $|B(x', R) \cap \Omega| \geq c R^n$. Since $B(x', R) \subset B(x, r)$, this proves $v \geq c r^n$.

Finally, if $\text{diam}(\Omega)/4 \leq r < 2 \text{diam}(\Omega)$, apply the established small-radius estimate at the same center with radius $r_0 = \text{diam}(\Omega)/8$. Since $r_0 > r/16$ and $B(x, r_0) \subset B(x, r)$, the same lower density follows. This completes the proof of Theorem 1.

New exponential-growth cases

Corollary 1. *The conclusion of Theorem 1 holds in each of the following cases.*

1. $\phi(t) = t^p e^{ct^\alpha}$, where $p > n$, $c > 0$, and $1 \leq \alpha < p/(p-1)$.

2.

$$\phi_\gamma(t) = e^{t^\gamma} - \sum_{j=0}^{\lfloor n/\gamma \rfloor} \frac{t^{\gamma j}}{j!}, \quad 1 \leq \gamma < 1 + \frac{1}{n}.$$

Proof. The first function dominates t^p , and $\log(1 + \phi(t)) = O(t^\alpha) = o(t^{p'})$.

Both displayed functions satisfy the nontriviality condition. Indeed, for the first one, monotonicity of the exponential factor gives

$$\frac{t^n}{\phi(t)} \int_0^t \frac{\phi(s)}{s^{n+1}} ds \leq \frac{1}{p-n}.$$

For the second, put $m = \lfloor n/\gamma \rfloor$ and $q = \gamma(m+1) > n$. The power series gives $\phi_\gamma(t) \geq t^q/(m+1)!$, while $\log(1 + \phi_\gamma(t)) = O(t^\gamma)$. Termwise integration of the same positive series shows that its C_{ϕ_γ} is at most $(q-n)^{-1}$. If $\gamma = 1$, then $q = n+1$ and $1 < q/(q-1)$. If $1 < \gamma < 1 + 1/n$, then $m = n-1$, $q = n\gamma$, and

$$\gamma < \frac{q}{q-1} \iff \gamma < 1 + \frac{1}{n}.$$

Apply Theorem 1 with exponent q . □

Limitations, novelty, and review recommendation

This is a partial answer. The proof does not cover the borderline $\log \phi(t) \asymp t^{p'}$, faster growth, or Young functions without a global lower power $p > n$. At the borderline, the exponent produced after (8) contains the extension and localization constants; nothing forces it below the summability threshold. Direct Orlicz–Poincaré and multilevel-cutoff attempts lead to the same obstruction.

A bounded search on 17 August 2026 covered the run indexes, the exact source wording, and the arXiv identifier, title, and authors. Close searches combined Orlicz–Besov extension, Ahlfors regularity, Trudinger, and $p/(p-1)$. The 2021 published source retains the subexponential hypothesis, and related 2025 work assumes doubling. No matching theorem was found. Novelty confidence is moderate.

Human review should focus on Lemma 1, the inverse- ϕ direction in (7)–(8), and the explicit recentring constants.

References

- [1] T. Liang and Y. Zhou, *Orlicz–Besov extension and Ahlfors n -regular domains*, Sci. Sin. Math. 51 (2021), 1993–2012; arXiv:1901.06186.
- [2] E. Parini and B. Ruf, *On the Moser–Trudinger inequality in fractional Sobolev–Slobodeckij spaces*, J. Anal. Math. 138 (2019), 281–300; arXiv:1607.07681.